Enhancing "stock-catalyst-historian": Data, Methods, and Honest Gaps

TL;DR

- **The highest-value free additions are regulatory disclosure feeds that cover your non-US universe** — exchange/FINRA short interest, FCA/ESMA net-short registers, IBKR borrow rates via iBorrowDesk, and catalyst feeds (ClinicalTrials.gov, USAspending). On methods, the best value-per-hour move is **meta-labeling on top of the D5 head** plus **conformal prediction** for honest per-pick intervals; the strongest econometric idea from Gemini (factor-residualized labels) is real but blocked because global point-in-time factor characteristics are paywalled.
- **Your biggest unnamed gap is survivorship bias, and it is unusually severe for an anomaly-event system** — anomalous movers are disproportionately likely to later delist, so training only on currently-listed symbols systematically hides the worst outcomes and inflates the D5 edge. Fix this before adding features.
- **Skip the résumé-driven complexity**: Hawkes processes, time-series foundation models, FastDTW pattern matching, and BMP/Kolari-Pynnönen event-study tests are not worth solo-dev hours for a tabular XGBoost trading system. Defer factor-residualized labels until you can source global point-in-time characteristics.

Key Findings

1. **Free data that covers the hard part (non-US, point-in-time-safe) exists and is under-exploited.** UK/EU short-selling registers (FCA, ESMA/national), exchange short interest (Nasdaq/NYSE), and IBKR borrow feeds fill the short-side gap that FINRA's paywalled float data left open. Catalyst APIs (ClinicalTrials.gov v2, USAspending, PatentsView) are free and event-shaped — exactly what a catalyst historian wants.
2. **The FINRA short-interest dead end has a genuine free workaround.** Consolidated short interest is published by the exchanges themselves (Nasdaq via FTP, NYSE semi-monthly file) and by FINRA's own bi-monthly equity short interest — this is the *official* twice-monthly data, distinct from the FINRA float endpoint behind the credential wall.
3. **Gemini's FINRA OTC/ATS (dark pool) suggestion is valid and free**, but the value is modest: it is a 2–4 week delayed weekly aggregate, US-only, and only weakly predictive at your 2-week horizon.
4. **Meta-labeling and conformal prediction are the two methods most worth your time.** Both sit cleanly on top of your existing XGBoost D5 head, are low-effort with mature libraries, and directly address your live problems (trade/no-trade gating; per-pick uncertainty).
5. **Factor-residualized labels (Gemini's strongest econometric idea) are blocked on data, not merit.** Per jkpfactors.com and Jensen, Kelly & Pedersen, "Is There a Replication Crisis in Finance?" (*Journal of Finance* 78(5):2465–2518, 2023), the dataset spans "153 characteristics/factors clustered into 13 themes... in 93 countries"; the long-short factor *returns* are free under CC BY-NC 4.0, but "Monthly stock-level characteristics data is

available via WRDS... Access requires a WRDS subscription." Ken French / AQR give you US and regional factor returns free but not per-stock point-in-time loadings.

6. **Survivorship bias is the dominant unnamed gap** and is worse for you than for a generic backtest, because your universe-defining event (an anomalous mover) correlates with later delisting.

Details

1. FREE DATA SOURCES — verified

Tier 1 — Build now (fills the non-US / short-side / catalyst gaps):

- **Exchange & FINRA short interest (official twice-monthly).** Nasdaq publishes a consolidated short interest report for all Nasdaq-listed securities via secure FTP, updated twice a month on the FINRA settlement schedule, with symbol, current/previous short shares, days-to-cover and split flags (per nasdaqtrader.com: "Nasdaq short interest is available by issue for a rolling 12 months and updated twice a month... released after 4:00 p.m, ET, on the dissemination date"). NYSE publishes the NYSE Group Short Interest File (semi-monthly, NYSE/Arca/American, history from 1988). This is the *official* short interest, sourced directly from the exchanges and FINRA, and it is genuinely free at the query level (per-security public pages; bulk via FTP). **Point-in-time safe** (settlement-date stamped). US-only. **Verdict: BUILD.** This is the correct replacement for the paywalled `short_interest_pct_float` FINRA endpoint. Note it is a snapshot on a lag, not real-time.
- **IBKR shortable-shares / borrow rates via iBorrowDesk.** Interactive Brokers' Securities Loan Borrow (SLB) system publishes indicative borrow rates, quantity available, and number of lenders, updated intraday; iBorrowDesk scrapes and exposes this with history. Coverage per IBKR's Securities Lending Dashboard is "for securities that trade in North America, Europe and Asia" (Orbisa-powered). This is a **borrow-cost / squeeze proxy that partially covers your global universe** — valuable because it is one of the few short-side signals that is not US-only. **Verdict: BUILD (via iBorrowDesk historical), but treat coverage as indicative, not a census.** Watch ToS on scraping; the SLB data itself is free to IBKR clients.
- **FCA net short position register (UK).** The FCA publishes short positions, but note a **major 2026 regime change** confirmed in FCA Policy Statement PS26/5, "Changes to the UK Short Selling Regime" (published 16 April 2026): "From 13 July 2026, the FCA will no longer disclose individual net short positions. It will instead publish anonymised aggregate net short positions (ANSPs) by company, combining individual positions reported at or above the 0.2% threshold... ANSPs will be disclosed from 12pm, two working days after the working day to which the positions relate." (Phase 1 commences 13 July 2026; Phase 2, bulk submission, 30 November 2026. Individual positions notified from 2021 remain archived.) **Verdict: BUILD** — free, ISIN-keyed, covers your UK universe, point-in-time-safe (daily working-day snapshots with dates). The move to aggregates actually simplifies ingestion.
- **ESMA / national short-selling registers (EU).** ESMA links to each national competent authority's public register; positions ⩾0.5% are disclosed. The ESMA Registers portal

offers machine-to-machine web services and a code-repository package for downloads. Coverage is fragmented across ~27 national sites but free and covers your EU universe. **Verdict: BUILD for the largest markets you trade (Germany, France, Nordics); defer the long tail.**

- **ClinicalTrials.gov API v2.** Free, no API key, JSON, global coverage (220+ countries, 500k+ studies; public-domain data). Queryable by sponsor company (`query.spons` / `query.lead`) and filterable by status (`filter.overallStatus`), sortable by last-update date to catch status changes. ~50 req/min informal limit; API v2 replaced the retired v1 in March 2024. **Point-in-time caveat: the API returns current status, not as-of-date history** — you must snapshot it yourself going forward to build a point-in-time series. **Verdict: BUILD for the biotech slice of your universe** — a clean, free, global catalyst feed. Harvard glama
- **USAspending.gov API.** Free, no authentication (official docs: "Endpoints do not currently require any authorization"), JSON, no formal rate limit. Query awards by recipient (use recipient UEI for stable identity). US-federal only. **30–90 day reporting lag** — good for confirmed historical contract catalysts, not same-day events (does not support CORS; call from a backend). **Verdict: BUILD as a low-priority catalyst feature for government-contractor names; the lag limits its edge.** Govconapi
- **OpenFIGI API v3.** Free, key optional but recommended (with key: 25 requests per 6 seconds and 100 jobs/request; without: 25/min and 10 jobs/request). Official docs: "The API is free and open to the public, though unauthenticated traffic will be subject to a lower rate-limit." Global, all asset classes; maps ticker/ISIN/CUSIP/SEDOL → FIGI. **V2 sunsets 1 July 2026 — build on v3. Verdict: BUILD** — this is infrastructure, not alpha: a stable cross-market security identifier to join Yahoo/FMP/EDGAR/exchange feeds, which directly reduces the symbol-mapping bugs that plague a 40-market universe. Openfigi

Tier 2 — Build if cheap / opportunistic:

- **CBOE free put/call ratios and volume.** CBOE publishes daily and historical index/equity/total put-call ratio CSVs free on its website (index/equity/total archives back to 1995–2006, plus post-2019 files via the Historical Options Data download). This is a **US market-level sentiment/risk gauge**, not per-stock options flow. **Verdict: BUILD the index/equity P/C ratio as a cheap regime/risk feature; there is no free per-stock options flow, IV surface, or gamma exposure — those require paid data.**
- **Companies House API (UK).** Free, API key required, 600 requests / 5 minutes (HTTP 429 on breach), JSON, read-only. Gives filing metadata, officers, PSC, filing history — but only ~40% of financial statements are digitised (the rest are PDFs). **Verdict: BUILD for UK filing-event detection (new filings as catalysts); skip it as a fundamentals source (coverage too patchy).**
- **SEC Financial Statement Data Sets (bulk XBRL).** Free quarterly bulk ZIP files of numeric data from the face financials of all US filers since 2009, presented "without change from the 'as filed' financial reports," updated quarterly (Financial Statement *and Notes* datasets monthly). This is a **free, genuinely point-in-time-safe alternative to**

FMP-premium point-in-time fundamentals for US names — because it is "as filed" with the filing date, you can reconstruct as-known-at-date fundamentals. **Verdict: BUILD** — the strongest free replacement for the FMP fundamentals you were considering paying for, for US filers. It will not help your non-US universe.

- **PatentsView PatentSearch API.** Free but **now requires an API key** — the legacy keyless API was discontinued 1 May 2025 and returns HTTP 410 ("Any requests made to api.patentsview.org will result in a 410 gone client error response"). 45 req/min per key, US patents only, queryable by assignee. **Viability caveat: rOpenSci reported in March 2026 that the API-key request link had been removed and long-term viability is uncertain. Verdict: DEFER** — verify key issuance is open before investing; US grants on a weekly lag are a weak catalyst signal relative to the integration risk.
- **Nasdaq Data Link (ex-Quandl) free datasets.** The 2021 transition removed many free datasets; most useful ones are now premium. **Verdict: SKIP as a general strategy; check individual free datasets opportunistically.**

Tier 3 — Paid, bundle into the FMP-premium decision:

- Per-stock options flow / implied volatility / gamma exposure (EODHD options, ORTEX, paid CBOE DataShop).
- Earnings-call transcripts: **there is no fully free, comprehensive, clearly-legal transcript API.** roic.ai's free tier is the best legal option (5 req/min, ~2 years/8 quarters history, global, no card) and Finnhub's free tier works for prototyping; API-Ninjas prohibits commercial use on free tier; FMP gates transcripts to its $139/mo Ultimate plan. **Scraping Motley Fool and Seeking Alpha violates their Terms of Use** (Motley Fool ToU explicitly prohibits scraping; Seeking Alpha is Cloudflare/paywall-gated) — ban and legal risk, not reliable for production. **Verdict: use roic.ai free tier for a FinBERT-tone transcript feature if you want it; otherwise treat transcripts as paid.** API Ninjas + 2
- Global point-in-time stock characteristics (JKP via WRDS) — see methods below.

2. ANALYSIS METHODS — value-for-effort for THIS system

High-value / low-effort — BUILD:

- **Meta-labeling (López de Prado) on the D5 head.** A secondary model that takes the D5 signal (and features) and predicts *whether to act* and *how big to bet*. This is the textbook fix for a system with one profitable horizon and marginal others: it separates side (D5 already gives you) from size, suppresses false positives, and improves precision without sacrificing recall. Mature, well-documented, sits directly on your existing pipeline. **Verdict: BUILD — highest value-per-hour method here.**
- **Conformal prediction (split/Mondrian) for per-pick intervals.** Replaces/augments your isotonic calibration with distribution-free prediction intervals that carry finite-sample coverage guarantees, assuming exchangeability. Mondrian (class/tier-conditional) conformal is ideal for giving each *tier* its own honest interval. Libraries (MAPIE and others) make this low-effort on tabular XGBoost. **Verdict: BUILD — pairs naturally with meta-labeling for bet sizing.** Caveat: exchangeability is violated by temporal drift, so use your existing purged/embargoed splits for the calibration set.

- **Deflated Sharpe Ratio + Probability of Backtest Overfitting.** These are post-hoc honesty checks tailored exactly to your risk: five heads × 100+ features × repeated re-analysis is textbook selection bias under multiple testing. The DSR (Bailey & López de Prado 2014) corrects the reported Sharpe for the number of trials, skewness and kurtosis; PBO (via combinatorially symmetric CV) estimates the chance your "winning" configuration is overfit. On the significance bar, Harvey, Liu & Zhu, "…and the Cross-Section of Expected Returns" (*Review of Financial Studies* 29(1):5–68, 2016) state: "it does not make any economic or statistical sense to use the usual significance criteria for a newly discovered factor, e.g., a t-ratio greater than 2.0... a newly discovered factor needs to clear a much higher hurdle, with a t-ratio greater than 3.0." **Verdict: BUILD — cheap to compute, directly bounds the credibility of your ~435bps edge claim.**
- **Volatility-scaled position sizing / fractional Kelly.** Size positions inversely to forecast volatility, cap at a fraction of Kelly to survive estimation error. Cheap, robust, and directly relevant given your £1.5k–2k minimum-viable-position constraint and small number of concurrent positions. **Verdict: BUILD — use GARCH or simple realized-vol for the vol estimate (see below).**
- **Cross-sectional ranking objective (LambdaMART / rank-IC loss).** Your system picks top-3 from ~1,400 daily — that is a ranking problem, not an absolute-return regression problem. Learning-to-rank research shows LambdaMART can materially improve rank-IC and portfolio Sharpe versus pointwise regression for exactly this top-of-book selection task (one cross-sectional-momentum study reports tripled Sharpe ratios and "LambdaMART delivers the best returns and ranking accuracy"), and recent work (LambdaRankIC) directly optimizes rank-IC with more overfitting robustness. XGBoost has native `rank:pairwise`/`rank:ndcg` objectives, so this is a **drop-in objective change**, not a new stack. **Verdict: BUILD/EXPERIMENT — high value, low-moderate effort; A/B against your regression head on realized cost-adjusted top-3 hit-rate.**

High-value / high-effort — DEFER or conditional:

- **Triple-barrier labeling vs fixed-horizon returns.** Path-dependent labels (take-profit / stop-loss / time barrier) are more realistic than fixed 5-day returns and are the natural substrate for meta-labeling. But they change your entire label pipeline across five horizons and complicate point-in-time bookkeeping. **Verdict: DEFER until after meta-labeling; then adopt triple-barrier for the D5 head specifically, since that is the one you trade.**
- **Regime detection (HMM / trend filter) to gate D5.** Gating the signal by market regime is well-motivated — the intraday-momentum literature finds a Markov-switching model "endogenously identifies two distinct regimes" and that a thresholded/regime-gated strategy "delivers higher returns than a strategy that is always active." But HMM stability and regime-labeling lookahead are real solo-dev traps. **Verdict: DEFER; start with a simple, transparent trend/vol filter (e.g., index above/below 200-day, VIX/put-call regime) before any latent-state model.**
- **Factor-residualized labels (Gemini's strongest econometric idea).** Training on factor-adjusted (residual) returns instead of sector-relative excess returns is genuinely better methodology and underpins Numerai's whole scoring philosophy (targets "neutralized

to a selection of standard factors: country, sector, beta, momentum, and size"). **The blocker is data, not merit:** you need per-stock, point-in-time factor loadings/characteristics for a *global* universe. As above, JKP factor *returns* are free (93 countries, CC BY-NC 4.0) but the stock-level *characteristics* are gated behind WRDS; Ken French and AQR give free factor returns for US/developed regions but not per-name point-in-time loadings. **Verdict: DEFER globally; optionally pilot on US names only using SEC XBRL-derived characteristics + French factors.** Meanwhile, adopt the cheaper cousin — **feature neutralization** (residualize features/predictions against sector/size/beta), which needs no external factor data and, in Numerai-style tests, "consistently reduced [volatility and maximum drawdown]... across all models." Jkpfactors

Not worth it — SKIP (with reasons):

- **Hawkes processes (multivariate marked, variational EM).** Event-clustering models are elegant but are a research project, not a feature; marginal lift over your existing event features does not justify the solo-dev cost. **SKIP.**
- **Time-series foundation models.** Overhyped for cross-sectional equity return prediction; heavy, opaque, and not competitive with tuned gradient-boosted trees on tabular financial features. **SKIP.**
- **FastDTW historical pattern matching.** Pattern-matching on price shapes is low-signal, high-maintenance, and duplicates what your 100+ features already encode. **SKIP.**
- **BMP test + Kolari-Pynnönen adjustment for event-study significance.** These are the right tools for an *academic event study* (testing whether abnormal returns are significant under event-induced variance and cross-sectional clustering). For a *trading* system whose truth is realized cost-adjusted PnL and out-of-sample IC, they answer a question you do not need answered. **SKIP — your DSR/PBO checks are the trading-relevant honesty tests.**
- **Interventional TreeSHAP for attribution.** SHAP capture at inference for per-pick attribution is already on your roadmap and is the right tool; interventional vs the default (Tree-path-dependent) SHAP is a refinement, not a new capability. **Keep your roadmap plan; the interventional variant is a nice-to-have, not a priority.**

FinBERT vs Gemini-2.5-Flash narrative/sentiment — a real look-ahead concern:

The look-ahead argument is legitimate and documented. Glasserman & Lin, "Assessing Look-Ahead Bias in Stock Return Predictions Generated By GPT Sentiment Analysis" (arXiv:2309.17322, 28 Sept 2023, GPT-3.5-Turbo at temperature 0), find: "In-sample (within the LLM training window), we find, surprisingly, that the anonymized headlines outperform, indicating that the distraction effect has a greater impact than look-ahead bias. This tendency is particularly strong for larger companies." Because Gemini-2.5-Flash has memorized historical outcomes, using it to score *historical* news for backtests risks contaminating your prized point-in-time correctness. FinBERT (free, local, ~50ms/doc on a small GPU; ProsusAI or FinBERT-tone variants) has no memorized-future problem and is reproducible. **Caveats:** FinBERT detects *tone*, not price impact, and a widely-cited practitioner test found it frequently wrong on "positive" headlines; but the academic FinBERT of Huang, Wang & Yang, "FinBERT: A Large Language Model for Extracting Information from Financial Text" (*Contemporary Accounting Research*, 2023) reports that "FinBERT's out-of-sample sentiment classification accuracy rate is 88.2%, whereas the LM dictionary, NB, SVM, RF, CNN, and LSTM rates are 62.1%, 73.6%, 72.6%, 71.9%, 75.1%, and 76.3%, respectively." **Verdict: BUILD FinBERT-tone as the *backtest-safe historical* sentiment feature on 8-K / transcript text, and keep Gemini for forward-only narrative generation (not historical feature construction).** Add entity anonymization to whatever LLM you use for scoring. This directly protects your house obsession with point-in-time correctness.

3. GAPS / FALLING SHORT

- **Survivorship bias (the dominant unnamed gap).** Your universe is defined by currently-listed symbols, so training data excludes delisted/acquired/bankrupt names. This is *worse* for an anomaly-event system than for a generic backtest: the very thing that puts a name in your scan — an anomalous price/volume event — correlates with later delisting (distressed spikes, buyout pops, fraud collapses). Excluding those hides your strategy's worst realized outcomes and inflates the D5 edge. On magnitude, Elton, Gruber & Blake, "Survivor Bias and Mutual Fund Performance" (*Review of Financial Studies* 9(4):1097–1120, 1996) estimate U.S. mutual-fund survivorship bias at 0.9% per annum; the larger 2–4% figures come from Rohleder, Scholz & Wilkens (2011) for hedge funds, whose higher attrition and steeper pre-closure losses widen the bias. For a system that *deliberately selects anomalous movers*, assume the upper end or beyond. Free/cheap mitigations: (a) **limit backtests to recent years** (bias grows with lookback); (b) capture delisting notices going forward (exchange delisting notices, SEC Form 25) so your *future* dataset is survivorship-free; (c) source delisted-name histories — no free CRSP equivalent exists, but Norgate Data (paid, retail-priced) and QuantRocket are the standard retail survivorship-free sources. **Fix this before adding any new features — it is a bigger threat to your reported edge than any missing data source.**
- **Corporate-action integrity via Yahoo adjusted close.** Known failure modes: Yahoo's back-adjustment silently rewrites history on new splits/dividends (breaking point-in-time), mishandles some spinoffs and special dividends, and its adjusted series is not survivorship-safe. **Recommendation: cross-check corporate actions against a second source; store raw prices + adjustment factors separately so you can reconstruct as-of-date prices rather than trusting a mutable adjusted series.**
- **Multiple-testing burden.** Five heads × 100+ features × repeated re-analysis is a false-discovery machine. **Bound it with DSR/PBO (above), a t-stat hurdle of ~3.0 (Harvey-Liu-Zhu), and a pre-registered feature/hypothesis log so you can count your true number of trials.**
- **Label noise from asynchronous global closes.** With 40+ markets and one daily scan, cross-market features mix stale and fresh prices (a Tokyo close is ~15 hours older than a New York close at scan time), contaminating both features and labels. **Recommendation: timestamp every price with its market-local close, compute cross-market features only from data strictly available at scan time, and consider market-cohort-specific label windows.** This is a point-in-time bug class you have not named.

- **Execution realism / signal decay.** Your signal is generated on delayed daily data and executed at next open. The overnight-drift and anomaly-decay literature shows anomaly returns accrue and reverse across intraday/overnight boundaries and that many effects have decayed since publication; a signal computed at close and traded at next open can lose a meaningful fraction of its edge. **Recommendation: measure your own close-to-next-open decay explicitly (compare theoretical close-execution vs realized next-open fills) and fold it into the cost model — you already model 7 cost components; signal decay is the missing 8th.**
- **Small-n rank instability at the top of the book.** Picking top-3 from ~1,400 daily means your recommendations sit in the noisiest part of the ranking. **Recommendation: measure rank stability (e.g., bootstrap the feature set / perturb inputs and see how often the top-3 persist), and prefer a top-k with k>3 filtered by the meta-label and conformal interval rather than a hard top-3.** The learning-to-rank objective (above) also directly improves top-of-book rank quality.
- **Other classic 2025–2026 retail-quant failure modes:** implementation/backtest-engine risk (the same strategy gives different numbers in different engines); regime-dependence of anomalies (an always-on strategy underperforms a gated one); and over-reliance on a single mutable free price source. All are addressed by the recommendations above.

Recommendations (staged)

Stage 1 — Fix integrity before adding anything (weeks):

1. **Quantify and mitigate survivorship bias.** Restrict backtests to recent years now; start capturing delisting events (SEC Form 25, exchange notices) so your forward dataset is survivorship-free; price a Norgate/QuantRocket subscription against the value of a trustworthy edge estimate. *Threshold to change course: if a survivorship-adjusted backtest cuts the D5 edge below ~200–250bps net of your ~110bps costs, re-scope deployment.*
2. **Add DSR + PBO** to your outcome tracking and a pre-registered trial log. *If PBO > 0.5 or the DSR is not significant, treat the edge as unproven.*
3. **Fix asynchronous-close contamination and add signal-decay as an 8th cost-model component.**

Stage 2 — Highest value-per-hour methods (weeks): 4. **Meta-labeling on the D5 head** + **conformal (Mondrian) intervals** for trade/no-trade and bet sizing. 5. **Volatility-scaled / fractional-Kelly sizing.** 6. **A/B a LambdaMART ranking objective** against the regression head on cost-adjusted top-3 hit-rate. 7. **FinBERT-tone** as the backtest-safe historical sentiment feature; restrict Gemini to forward-only narrative; anonymize entities.

Stage 3 — Highest value-per-hour data (weeks): 8. **OpenFIGI v3** as the cross-market identifier backbone (reduces 40-market join bugs). 9. **Exchange/FINRA short interest + iBorrowDesk borrow** (short-side signal that partially covers non-US). 10. **FCA ANSP + ESMA/national short registers** for your UK/EU names (mind the 13 July 2026 FCA switch to anonymized aggregates). 11. **SEC XBRL Financial Statement Data Sets** as the free point-in-time US fundamentals source (defer/avoid paying FMP premium for US point-in-time). 12. **ClinicalTrials.gov v2** for the biotech slice; **CBOE put/call** as a market-regime feature.

Stage 4 — Defer/conditional: 13. Triple-barrier labeling (after meta-labeling); a simple regime gate before any HMM; factor-residualized labels only if you can source global point-in-time characteristics (pilot US-only with SEC XBRL + French factors).

Skip list (Gemini suggestions that don't survive scrutiny):

- **Hawkes processes** — research project, marginal lift, not a feature.
- **Time-series foundation models** — overhyped, not competitive with tuned GBMs on tabular data.
- **FastDTW pattern matching** — low-signal, high-maintenance, duplicates existing features.
- **BMP + Kolari-Pynnönen event-study tests** — answer an academic question, not a trading one; DSR/PBO are the trading-relevant checks.
- **EODHD options API** — paid, conflicts with the free constraint (bundle into a paid decision if per-stock IV ever becomes a priority).
- **PatentsView** — now key-gated with uncertain 2026 viability and weak signal; defer.
- **Netrows** — not a recognized reliable free source; skip pending verification.

Caveats

- **The ~435bps D5 edge is very likely overstated** until survivorship bias and multiple-testing are corrected; treat it as an upper bound.
- **Global point-in-time factor characteristics remain the key unlock you cannot get free** — the JKP data's stock-level layer needs WRDS; this genuinely caps how far the factor-residualization idea can go for a retail solo dev.
- **Short interest and regulatory short registers are snapshots on a lag** (twice-monthly; FCA ANSP ~2 working days from 13 July 2026; OTC/ATS 2–4 weeks) — useful as slow-moving features, not timing signals.
- **ClinicalTrials.gov, Companies House, and exchange feeds return current state, not as-of-date history** — you must snapshot them forward to preserve point-in-time correctness; do not trust their present values in historical backtests.
- **FinBERT detects tone, not price impact**, and can be wrong on positive headlines; validate any sentiment feature against realized returns before trusting it.
- Several sources (iBorrowDesk, Motley Fool/Seeking Alpha) involve scraping that may violate terms of service — prefer official feeds where they exist.